



Kseniia Harshina, Marie Biedermann, Rachel Gorden

Gamifying Data Science through Implementing Interactive Fiction in Twine

Who are we?

Marie

- GSE Student
- Made a Video Game for FH Kärnten
- Collaboration with Mi'pu'mi Games Vienna
- Wants to work as a Game Designer

https://vulvarine.itch.io/

Kseniia

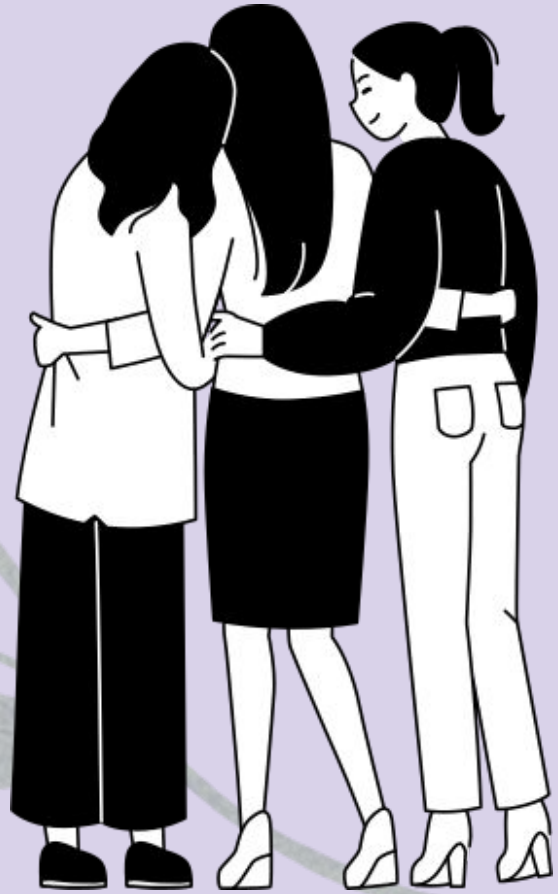
- Is pursuing a PhD in Games
- Worked in AI research for two years
- Likes to promote intersectionality in games
- Definitely doesn't play the same game every day

https://kseniaharshina.itch.io/

Rachel

- comes from humanities
- MA in Game Studies and Engineering
- likes game jams
- loves Stardew Valley

https://rachelgo.itch.io/



Who are you?

What field do you work in/ what are you studying?

What is your relationship to games?

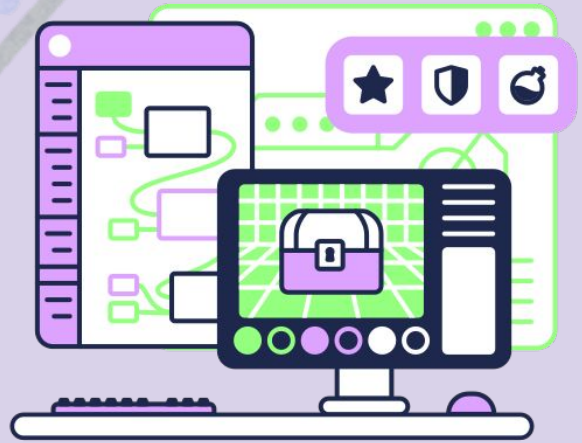
What is your motivation for joining today's workshop?



What we will do today

Why (and how) combine games and data science?

1) Why (and how) combine education and games?
2) What is Twine and how can we use it?
3) hands-on Game Jam-like workshop:
   a) create your own small games!



Video Games and Data Science

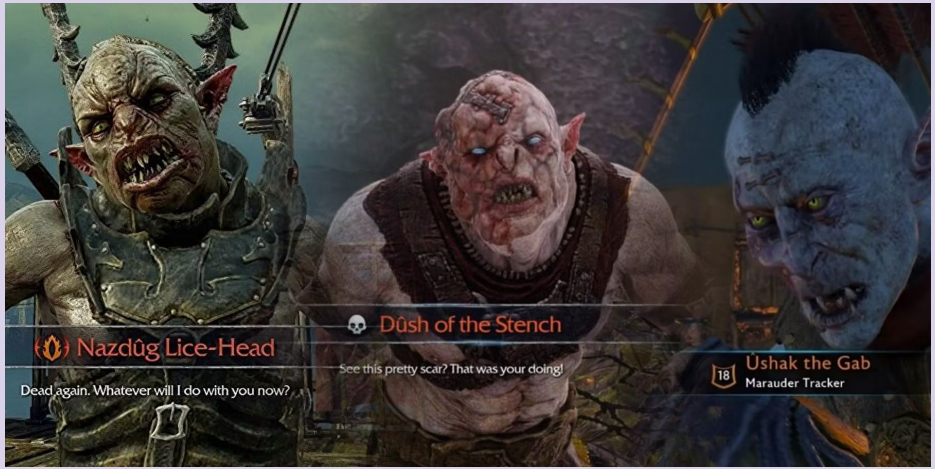
- Data-driven Gameplay

Data-driven gameplay refers to the use of **real-time data** and **player feedback** to dynamically alter game environments, challenges, and narratives.

It allows developers to tailor gaming experiences to individual player behaviors and preferences, making games more engaging and responsive.

Video Games and Data Science

- Data-driven Gameplay



Nemesis System

AI Director



Video Games and Data Science

- Predictive Analytics

Predictive analytics involves using **historical** and **real-time data** combined with statistical algorithms and machine learning models to predict future player behaviors, preferences, and game outcomes.

Video Games and Data Science

- Predictive Analytics



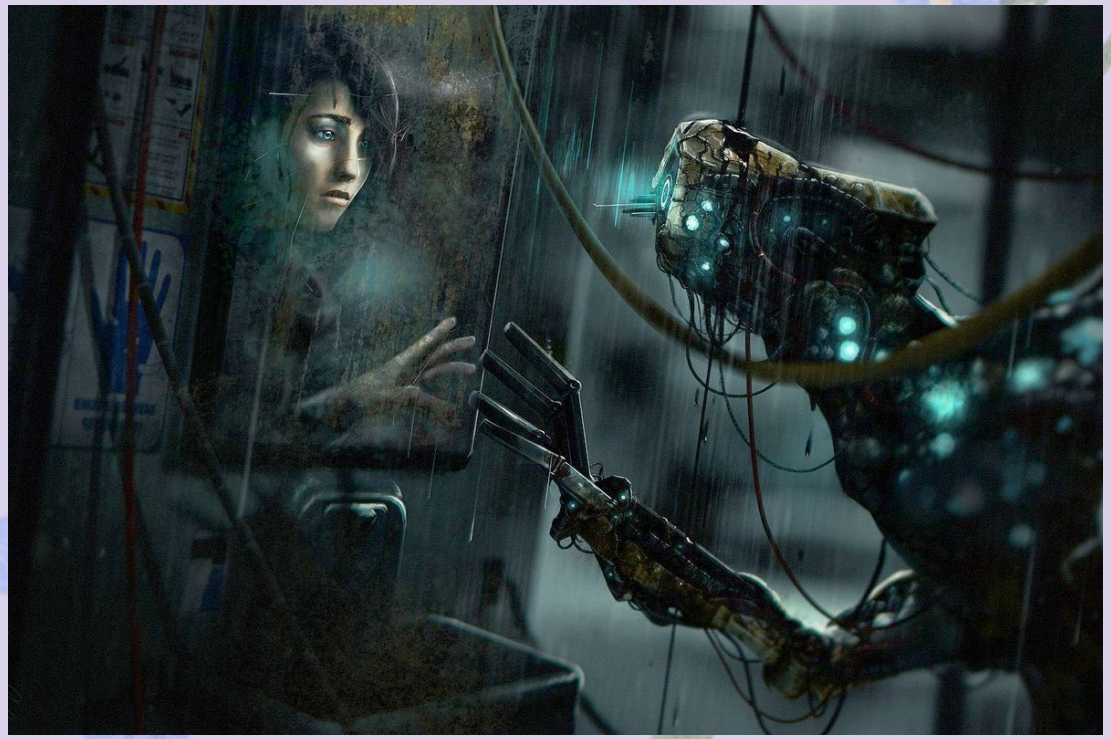
Valorant: Matchmaking, Balancing



Mobile games: player retention and monetization

Video Games and Data Science

- AI and Future Possibilities





Video Games and Data Science

- Data Analytics for Indie Game Developers



The Itch, the Game and the Ugly: How to Incorporate Data Analytics into the Game Development Process to Boost Player Engagement on Itch.io

Video Games and Education

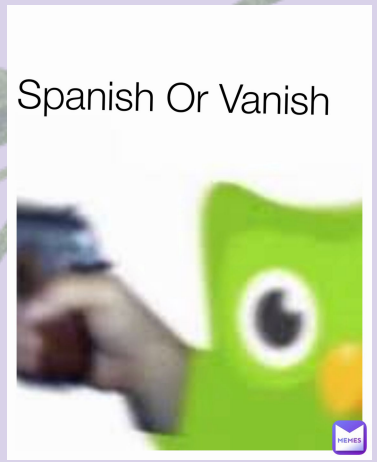
- What is Gamification?

Gamification is the process of applying **game-design elements** and game principles in non-game contexts. This can include *points, levels, challenges, rewards, and player engagement mechanics* to motivate participation and engagement.

Video Games and Education

- What is Gamification?







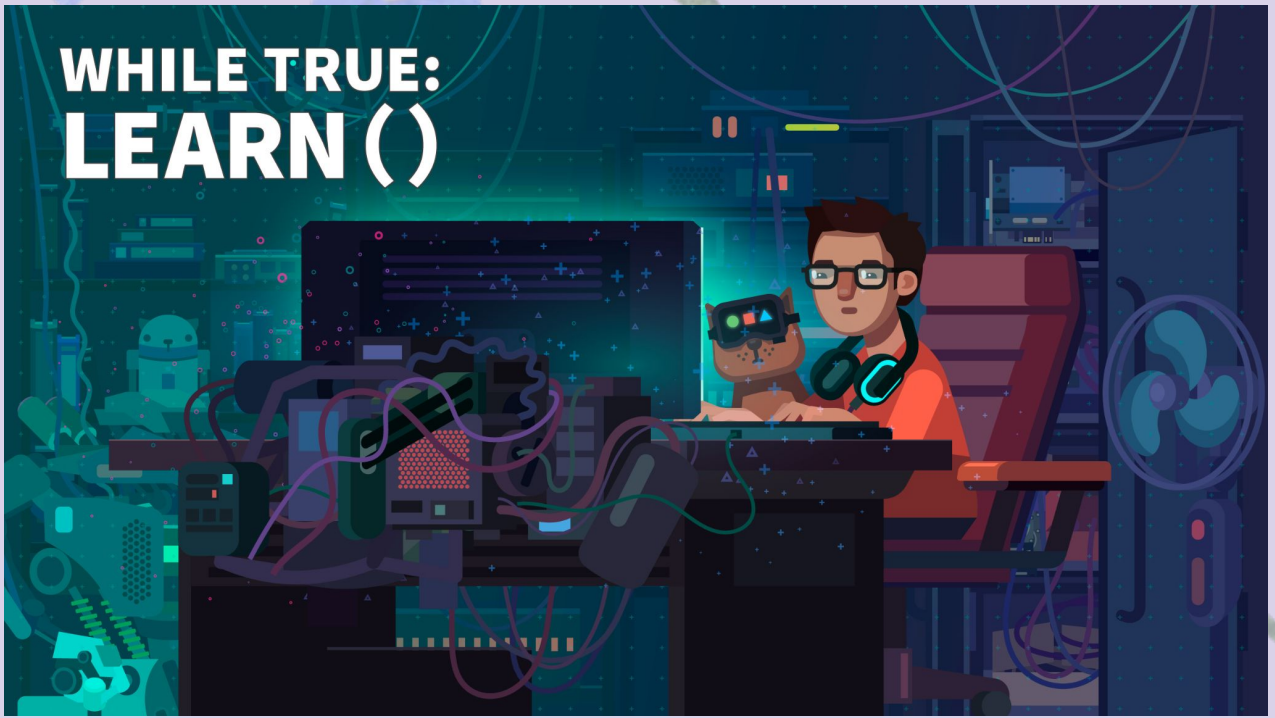
Video Games and Education

- Gamification: Examples



Video Games and Education

- Gamification: Examples



Video Games and Education

- Gamification: Examples



Twine

- interactive storytelling tool



Examples of games made in Twine:

- https://w.itch.io/end-of-the-world
- https://philome.la/Proelectioneer/americas-refugee-policies/index.html
- https://tan-tan.itch.io/do-not-go-into-the-gentle-night

Twine 101

- **Creating** and **exporting** a project
- Creating different story-structures
- Including *variables*
- Changing the **look** of the game
- Adding assets to your game

- Some ideas to get you started…



Creating a new project in Twine (Harlowe)



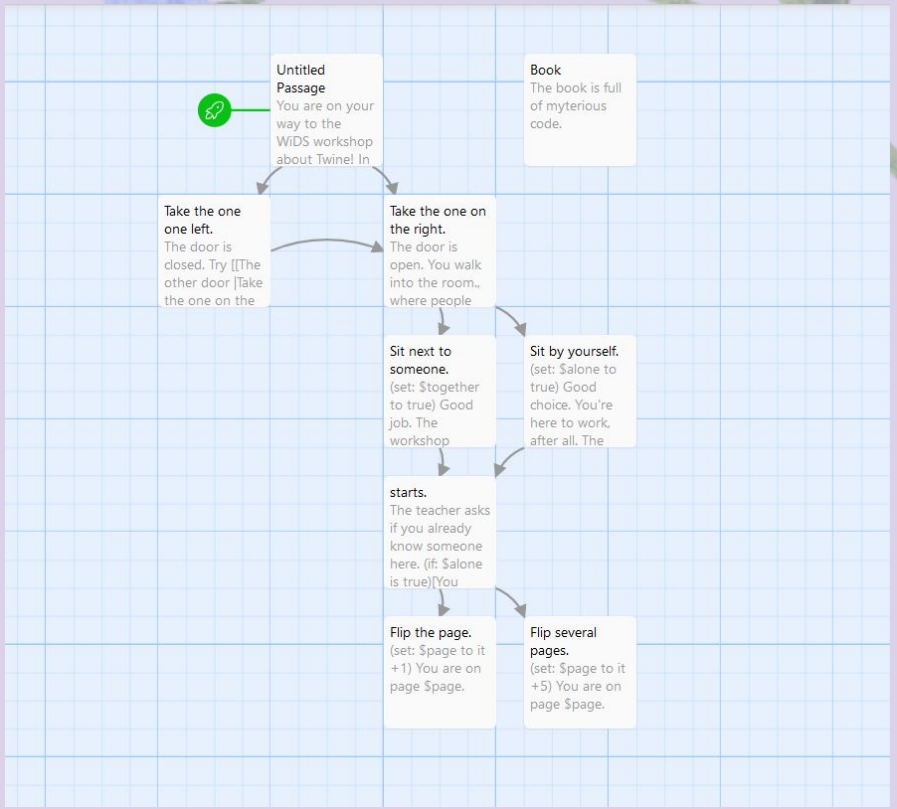
go to https://twinery.org/

Basics:

“I wanna go **[[here]]”** → brings you to the next passage

Try it: write a text in your start passage and add a link to the next passage of the story!

Non-linear storytelling in Twine



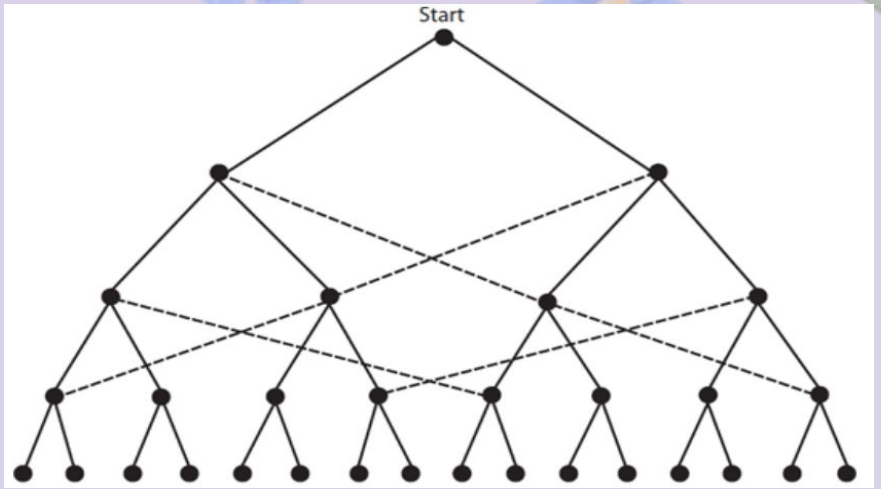
Try it:

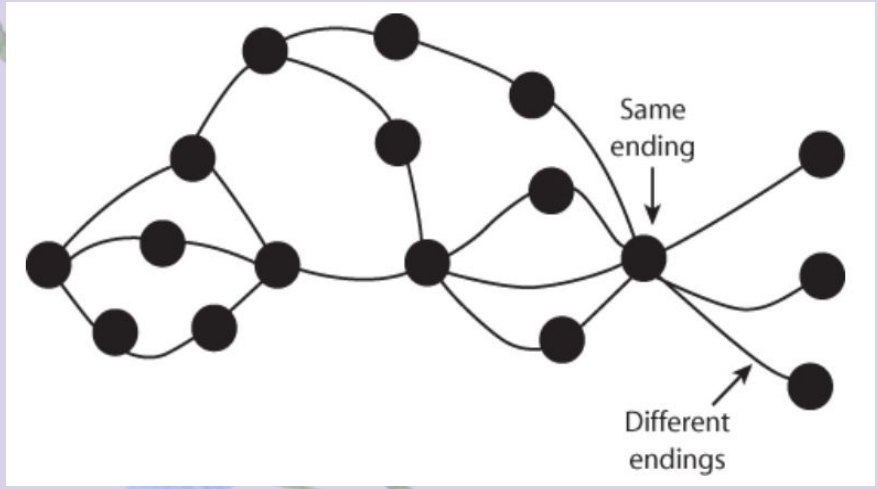
add a **choice** to your game that leads to different passages

example:

"Should I go **[[left]]** or **[[right]]**?"

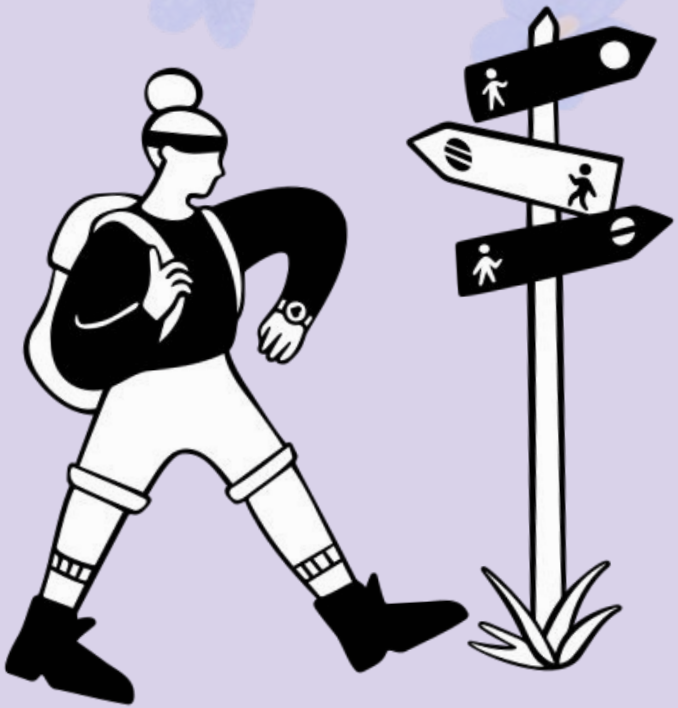
Non-linear Storytelling





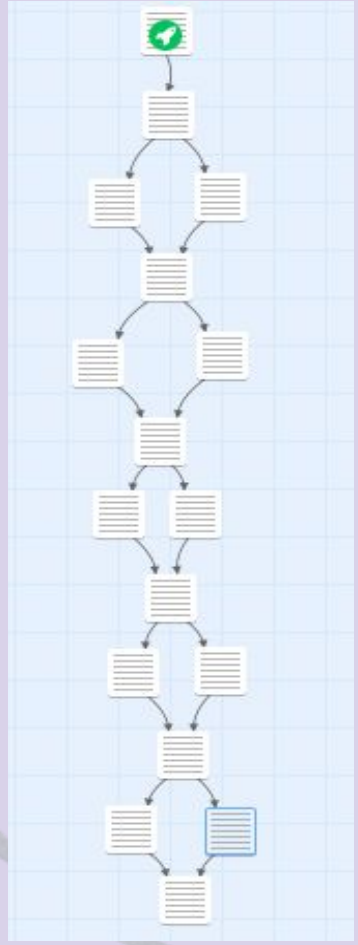
Source: Ryan, Marie-Laure. 2001. *Narrative as Virtual Reality: Immersion and Interactivity in Literature and Electronic Media.* Baltimore and London: John Hopkins University Press.

Non-linear Storytelling

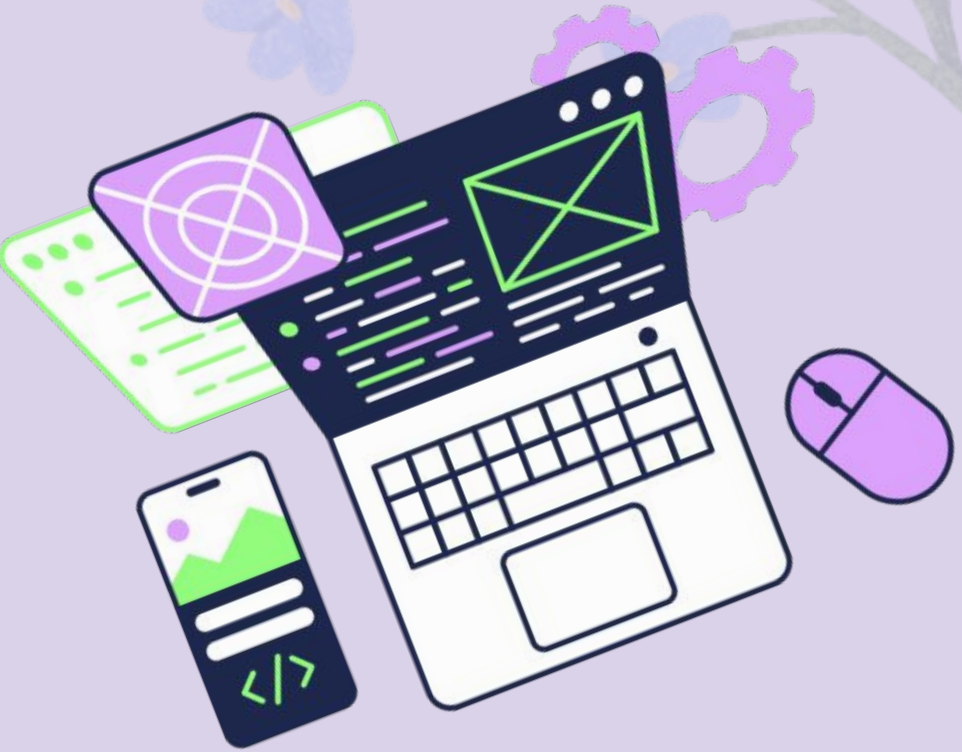


Try it:

Have all passages in your story lead back to the same passage!



Gamifying your Story with Variables



- numbers, strings (text), booleans (true/false)...
- stores data throughout the game
- data can be changed and/or displayed at any point

Why use Variables?

- **to remember** choices that players made earlier in the story
  - this can open up **more/less possibilities** in the game
  - **Example:** a boolean variable named $I_know_this_goose is set to **true** when the player meets a **goose**. Later in the game, players who know the goose (for whom this boolean is set to true) are offered to join the goose on an adventure.
- to count the player's score (with or without showing it)
- to let players collect items
- to let players reach certain endings of a story
- etc.



Storing Variables

booleans

(set: $example to true)



numbers

(set: $count to 1)

(set: $count to it +1)

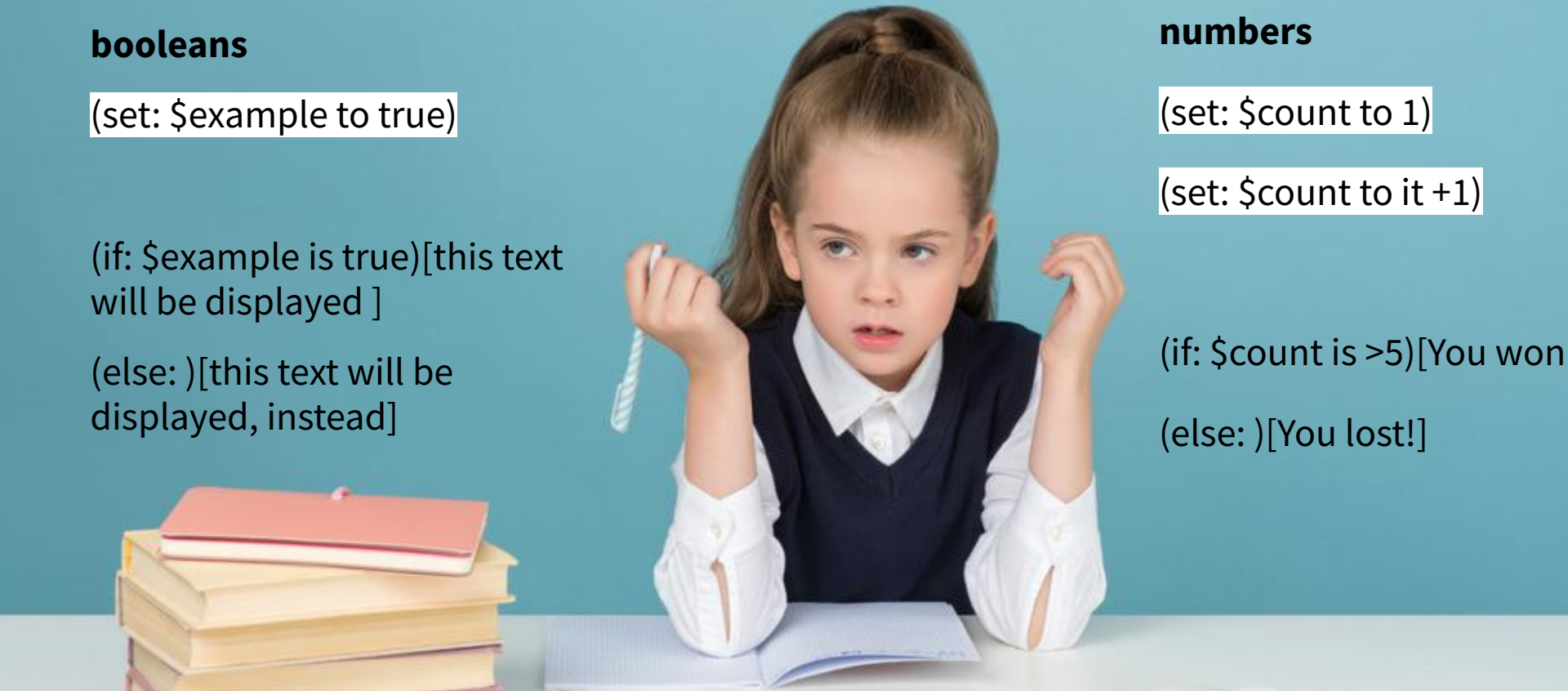
Using Variables

booleans

(set: $example to true)

(if: $example is true)[this text will be displayed]

(else:)[this text will be displayed, instead]



numbers

(set: $count to 1)

(set: $count to it +1)

(if: $count is >5)[You won

(else:)[You lost!]

Variables

Try it:

1. Create a variable storing a number in a passage
2. Give the player a choice
3. For each choice, change the variable to a different number
4. Display the variable in your story



Example:

(set: $geese to 0)

I wanna have two geese.

(set: §geese to it +2)

or

I'd rather have three geese.

(set: §geese to it +3)

I now have $geese geese!

Styling your game

Styling **text** (headings, emphasis, italics, …):

https://twinery.org/cookbook/markup/harlowe/harlowe_markup.html

- here you can also download the example project!



Styling your game

```
:: StoryTitle
Style Markup in Harlowe
:: Start
//Italics//
"Boldface"
~~Strikethrough text~~
*Emphasis*
**Strong emphasis**
Super^^script^^
``[[Escaped double square brackets]]``
#Level 1 heading
##Level 2 heading
###Level 3 heading
####Level 4 heading
#####Level 5 heading
######Level 6 heading
```

```
* Bulleted item
    *   Bulleted item 2
 ** Indented bulleted item
 0. Numbered item
   0. Numbered item 2
 0.0. Indented numbered item
 ==> This is right-aligned
  =><= This is centered
 <==> This is justified
<== This is left-aligned (undoes the above)
===><=This has margins 3/4 left, 1/4 right
  =><=====This has margins 1/6 left, 5/6 right.
|== This is in the leftmost column, which has a right margin of about 2 letters wide.
```

```
    =|||=
This is in the next column, which has margins of 1 letter wide. It is three times as wide as the left column.
 =====||
This is in the right column, which has a right margin of about 5 letters wide. It is twice as wide as the left column.
  |==|
This text is not in columns, but takes up the entire width, as usual.
```

Styling your game

Using **Google Fonts**:

https://twinery.org/cookbook/googlefonts/harlowe/harlowe_googlefonts.html

★ here you can also download an Example project

https://fonts.google.com/

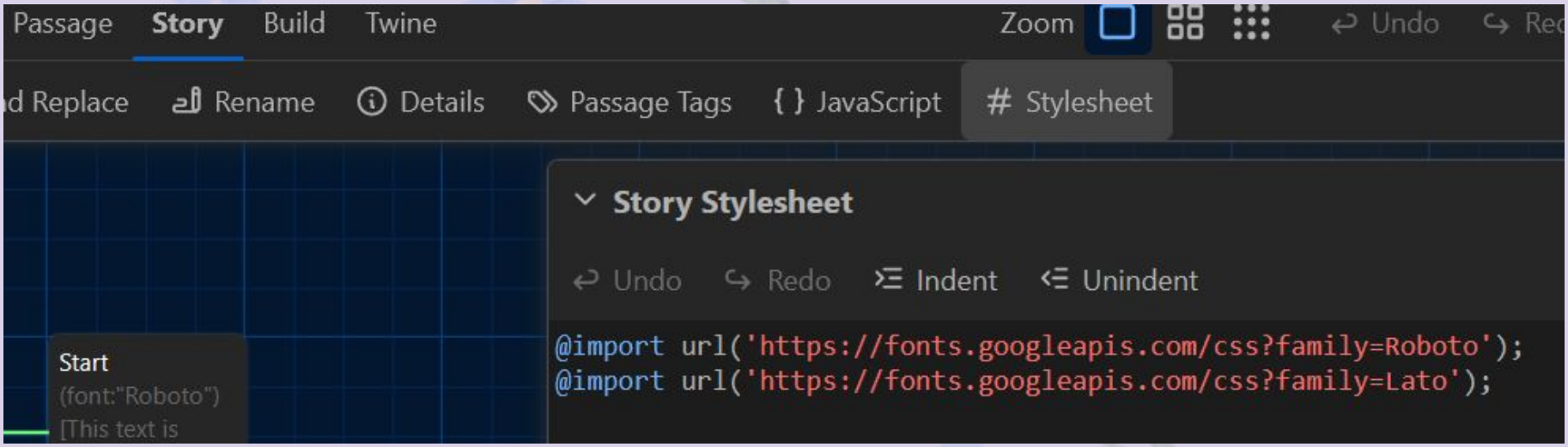
★ Here is a list of Google Fonts (not all are usable I think)

★ :: StoryTitle
  - Harlowe: Google Fonts

★ :: StoryStylesheet[stylesheet]
  - @import url('https://fonts.googleapis.com/css?family=Roboto');
  - **!!Import this link with the name of the fontfamily!!**

★ :: Start
  - (font:"Roboto")[This text is styled by a Google Font]

Styling your game



Adding Assets to your Game



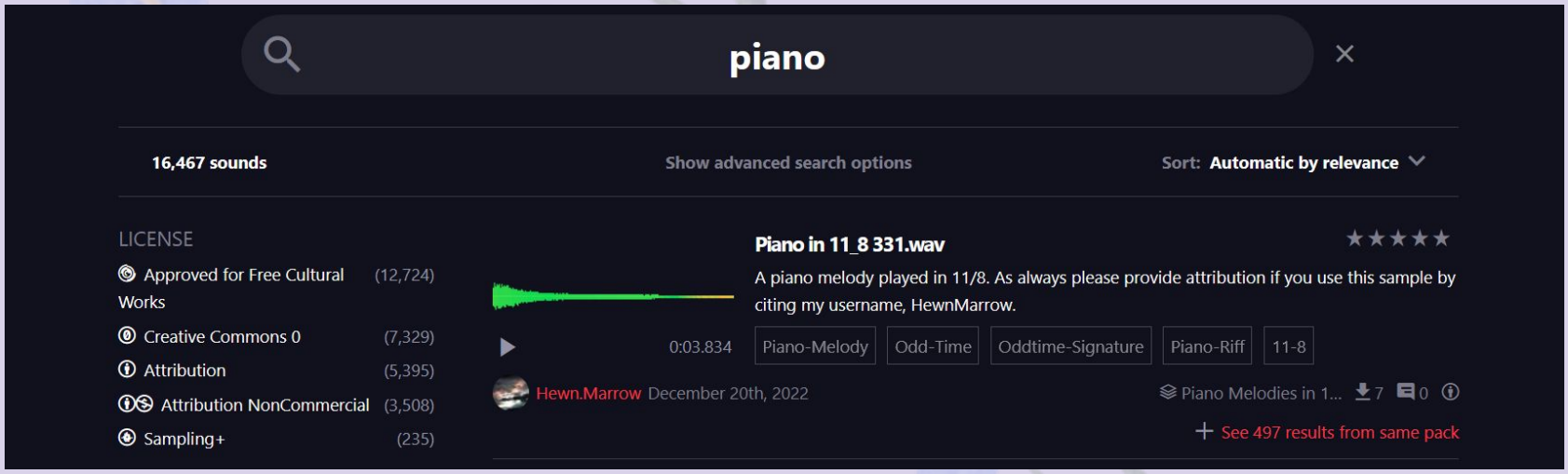
Adding **audio**:

https://twinery.org/cookbook/audio/harlowe/harlowe_audio.html

Adding **images**:

https://twinery.org/cookbook/images/harlowe/harlowe_images.html

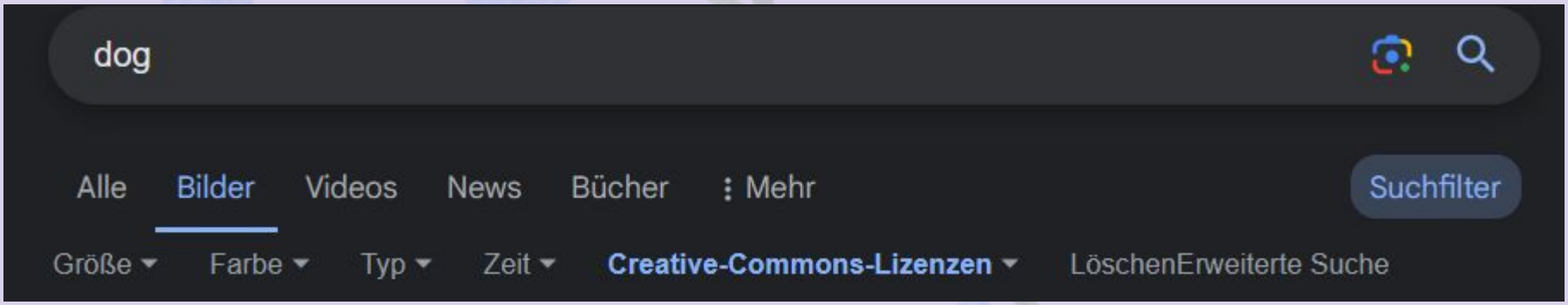
Adding Assets to your Game



Freesound.org: Login - type in wished sound/music (piano here) - click on:

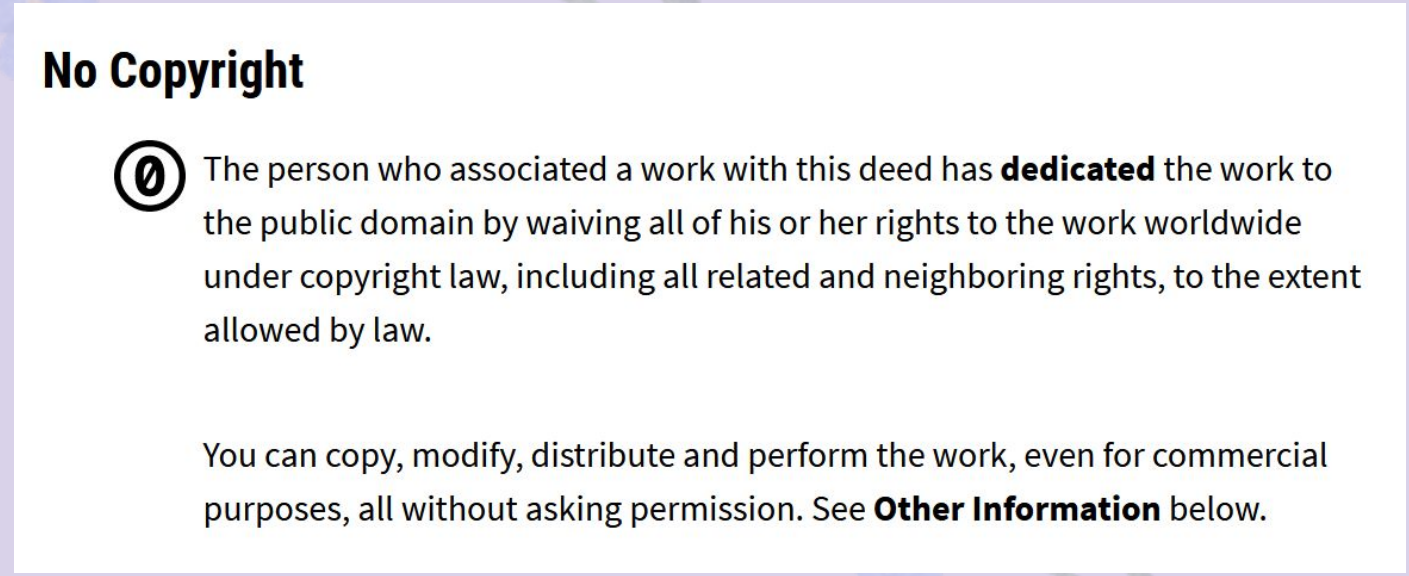
Creative Commons 0

Adding Assets to your Game



Google: Bilder - Nutzerrechte - Cretive-Commons-Lizenzen

Adding Assets to your Game



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Short Break



Let’s get Creative!

Ideas to get you started:

- present a data set in the form of a story (https://www.kaggle.com/datasets)
- make a quiz that tests players’ knowledge (and keeps score)
- create a non-linear story following your day-to-day tasks at work to show what you do

Example topics that your game can revolve around:

- artificial intelligence
- statistics
- coding



Pitch Your Idea

If you want to pitch us your idea

THE TIME IS NOW!

30 SECONDS

Resources for Game Development with Twine

Documentation of how to use Twine: https://twinery.org/cookbook/

Harlowe manual and documentation: https://twine2.neocities.org/

free to use assets, audio, etc:

Playtest!

- Send us your HTML File!
  - funwithfrienz@gmail.com
- Do we want to create an itch.io page to collect the games to showcase?

Icons by https://icons8.com/